## 1. Source Code

- **Definition**:

The **original code written by the programmer** in a high-level programming language like Java, C, Python, etc.

- **Written by**: Humans
- **Readable by**: Humans
- **Example**:

```
public class Hello {
   public static void main(String[] args) {
      System.out.println("Hello World");
   }
}
```

- **File extension (Java)**: .java

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## 2. Bytecode (Java Specific)

- **Definition**:

**Intermediate code** generated by the **Java Compiler (javac)** after compiling the source code.

- **Readable by**: Java Virtual Machine (JVM), **not humans**
- **Platform-independent** (runs on any OS with JVM)
- **File extension**: .class file

- **Example process**:

javac Hello.java → generates → Hello.class

This .class file contains bytecode.

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## 3. Machine Code (Object Code)

- **Definition**:

**Low-level binary code** that the actual **hardware (CPU)** can understand and execute.

- **Generated by**:

- JVM (in Java, using **Just-In-Time (JIT)** compiler)
- Or directly by compiler in C/C++

◆ **Readable by**: Machine only (binary form: 0s and 1s)

◆ **Platform-dependent**

◆ **Example** (not human readable):

10111000 00001101 11001001 ...

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## Java Execution Flow Summary

1. Write Source Code (.java)

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2. Compile with javac → Bytecode (.class)

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3. JVM interprets or compiles → Machine Code (executed by CPU)

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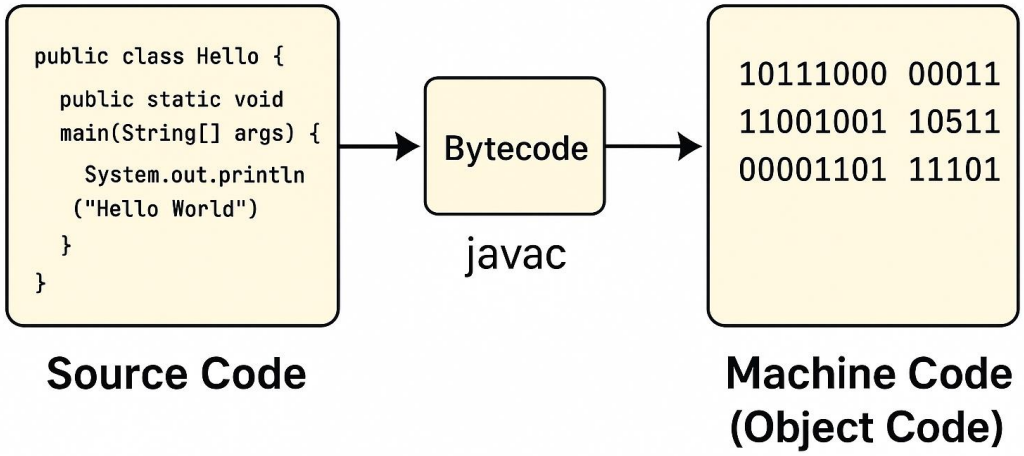
**Key Differences Table:**

| Feature | Source Code | Bytecode | Machine Code |
|---|---|---|---|
| Written By | Programmer | Java Compiler | JIT or Native Compiler |
| File Format | .java | .class | Binary (no extension) |
| Readable By | Humans | JVM | CPU |
| Portability | Platform-independent | Platform-independent | Platform-dependent |
| Usage | Write logic | Intermediate representation | Final execution code |

# Source Code, Bytecode, Machine Code (Object Code)




**Source Code**

**Machine Code (Object Code)**